OÉ Gaillimh
NUI Galway

Semester II Examinations 2010/ 2011

Exam Code(s)	1MF1, 3BA1, 3BS9, 3EL2, 4BS2, 4CS2,
Exam(s)	3rd & 4th Science, H. Dip. (Appl. Sc), M.Sc. Software Design & Development
Module Code(s)	MP307
Module(s)	Modelling II
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor D.C.G. McKeon
Internal Examiner(s)	Professor M. Destrade Dr P.T. Piiroinen
***Instructions*:**	For full marks answer question 1 in Section A **and** TWO questions in Section B correctly. All questions carry the same weight.
Duration	2hrs
No. of Pages	5
Department(s)	Applied Mathematics
Course Co-ordinator(s)	Dr P.T. Piiroinen

Requirements:

MCQ	
Handout	
Statistical/ Log Tables	Yes
Cambridge Tables	
Graph Paper	
Log Graph Paper	
Other Materials	

Section A - Answer all questions

1. (a) Consider a queue and let

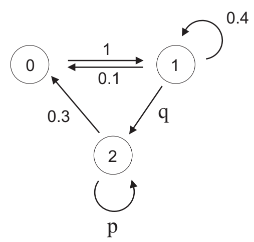
$$p_k(t) = \frac{(\alpha t)^k}{k!} e^{-\alpha t}, \quad \alpha > 0 \quad \text{(Poisson distribution)}$$

be the probability for k arrivals in time t.

(i) Calculate the avarage number of arrivals per unit time.

(ii) Calculate the variance and standard deviation?

(b) Consider the following Markov Process with transition probablities shown.



(i) Find the transition matrix P and the probabilities p and q.

(ii) Is the Markov process ergodic? Explain your answer.

(iii) Find the equilibrium probabilities.

(c) Consider a model for a non-fatal disease in a population P given by

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI, \quad S(0) = S_0, \ \alpha > 0, \\ \frac{dR}{dt} &= \beta I, \quad R(0) = 0, \ \beta > 0,\end{aligned}$$

where S, I and R respectively correspond to susceptible, recovered and infected groups of the population. Let $\rho = \beta/\alpha$ and show that the relation between the groups S and R can be described by

$$\begin{aligned}S(R) &= S_0 e^{-\frac{R}{\rho}}, \\ R(S) &= -\rho \ln\left(\frac{S}{S_0}\right).\end{aligned}$$

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= x_2,\\ \frac{dx_2}{dt} &= x_1 + x_2 + b_1 u\\ y &= x_2\end{aligned}$$

where x_1 and x_2 represent the state, $u \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b_1 \in \mathbb{R}$ is the control parameter.

(i) Is this system observable? Explain your answer.

(ii) Is this system reachable? Explain your answer.

Section B - Answer any **two** questions.

2. Consider a **two-state** queue (eg. a single telephone line) with arrival and service patterns described by Poisson processes with parameters α and β, respectively. Let

$$p_k(t) = \text{Prob}(k \text{ customers queuing at time } t), \quad k = 0, 1.$$

   (a) Sketch the corresponding transition diagram.

   (b) Show that the probabilities $p_0(t)$ and $p_1(t)$ evolve according to the differential equations

$$\begin{aligned} \frac{dp_0(t)}{dt} &= -\alpha p_0(t) + \beta p_1(t), \\ \frac{dp_1(t)}{dt} &= \alpha p_0(t) - \beta p_1(t). \end{aligned}$$

   (c) Calculate the equilibrium probabilities π_0 and π_1 for the two-state queue.

   (d) Consider a single telephone line and let 5 minutes be the average time for a call to arrive and let the average length of a call be 4 minutes. What is the probability that the phone line is busy?

3. Two competing species with population sizes P_1, P_2 evolve according to the following differential equations

$$\begin{aligned} \frac{dP_1}{dt} &= 6P_1 - 2P_1^2 - 2P_1P_2, \\ \frac{dP_2}{dt} &= 4P_2 - 2P_1P_2 - 2P_2^2. \end{aligned}$$

   (a) Give an interpretation for the various terms in these equations.

   (b) Find the equilibrium points for P_1, P_2.

   (c) Show that for any initial non-zero populations $P_1(t) \to 3$ and $P_2(t) \to 0$, as $t \to \infty$. What is the interpretation of this result?

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_1 + c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0, p_1, p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the sum of the concentrations of the drug in compartments 1 and 2.

(a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.

(b) Investigate for what values of p_0, p_1 and p_2 this system is observable.

(c) Design a control function

$$u = -Kc + k_r r, \quad \text{for} \quad K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 2$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -2$ and $\lambda_2 = -1$.